



Foundations of Finance



Equity Valuation II



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Partial Review of the Last Class

- ▶ A share trades at \$48. You expect a \$4 dividend and a \$52 price next year, and $r = 8\%$. Is it cheap or expensive?
- ▶ JPMorgan's perpetual preferred pays \$1.20 a year and trades at \$15. What return does it offer?
- ▶ $D_1 = \$2$, $g = 4\%$, $r = 8\%$. What is the share worth, and what happens if g falls to 3%?
- ▶ If the formula is the same, why is a share so much harder to value than a bond?

Today

- ▶ Finishing Gordon: what the price says about the return
- ▶ Valuation ratios — P/D and P/E seen through Gordon
- ▶ Where growth comes from: $g = b \times \text{ROE}$
- ▶ When growth creates value, and when it destroys it
- ▶ The present value of growth opportunities (PVGO)
- ▶ Growth that does not last for ever: the two-stage DDM

Expected Stock Returns

- ▶ Under GGM, $P_0 = V_0 = \frac{E[D_1]}{(r-g)}$:

$$r = \frac{E[D_1]}{P_0} + g$$

Dividend
yield

Capital gain

- ▶ Ex: the stock pays investors its required return of 8% with a dividend yield of $\frac{2}{50} = 4\%$ plus a capital gain of 4%
- ▶ If the stock is under-valued, $P_0 < V_0$, then its expected holding period return is too high:

$$E[HPR] = \frac{E[D_1]}{P_0} + g > r$$

- ▶ Similarly, $E[HPR] < r$ if it is over-valued

The Gordon Model and the IBEX

- ▶ **Two very different ways for a share to deliver the same r :**
 - ▶ a high dividend yield and little growth — the Spanish banks and utilities that dominate the IBEX
 - ▶ almost no dividend, with the whole value sitting in g — US technology
- ▶ A high dividend yield is not by itself a cheap share. It is a share whose value is in D_1 rather than in g
- ▶ It is also why the IBEX 35 and the IBEX 35 con dividendos drift so far apart over a decade
- ▶ **Two shares at \$100, both with $r = 8\%$: A pays \$6 and grows 2%, B pays \$1 and grows 7%**
 - ▶ lose one point of growth and A falls to \$85.71 (−14%), B falls to \$50.00 (−50%)

Valuation Ratios

- ▶ Valuation ratios scale stock prices by some observable metric, which allows for comparison across firms
 - ▶ Ex: price-dividend (P/D), price-earnings (P/E), price-book (P/B), price-sales (P/S)
- ▶ We can interpret them using GGM. If $P_0 = V_0$, then

Price to trailing dividend	→	$\frac{P_0}{D_0} = \frac{(1+g)}{r-g}$	and	$\frac{P_0}{E[D_1]} = \frac{1}{r-g}$	←	Price to forward dividend
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- ▶ Investors often value a stock by assigning it the same valuation ratio or “multiple” as another, comparable stock
 - ▶ Implicitly assumes that the two stocks have the same R and g

Using GGM to Study Valuation Ratios

- ▶ Price-dividend ratio
- ▶ If GGM applies then:

$$\frac{P_0}{D_0} = \frac{1 + g}{r - g}$$

- ▶ Define the “earnings retention ratio” or the “plow-back ratio” b as the share of earnings that are retained in the company.
- ▶ The dividends paid out are then given by $D_0 = (1 - b)E_0$.
- ▶ If the GGM applies then the Price-Earnings ratio is given by:

$$\frac{P_0}{E_0} = \frac{(1 + g)(1 - b)}{r - g}$$

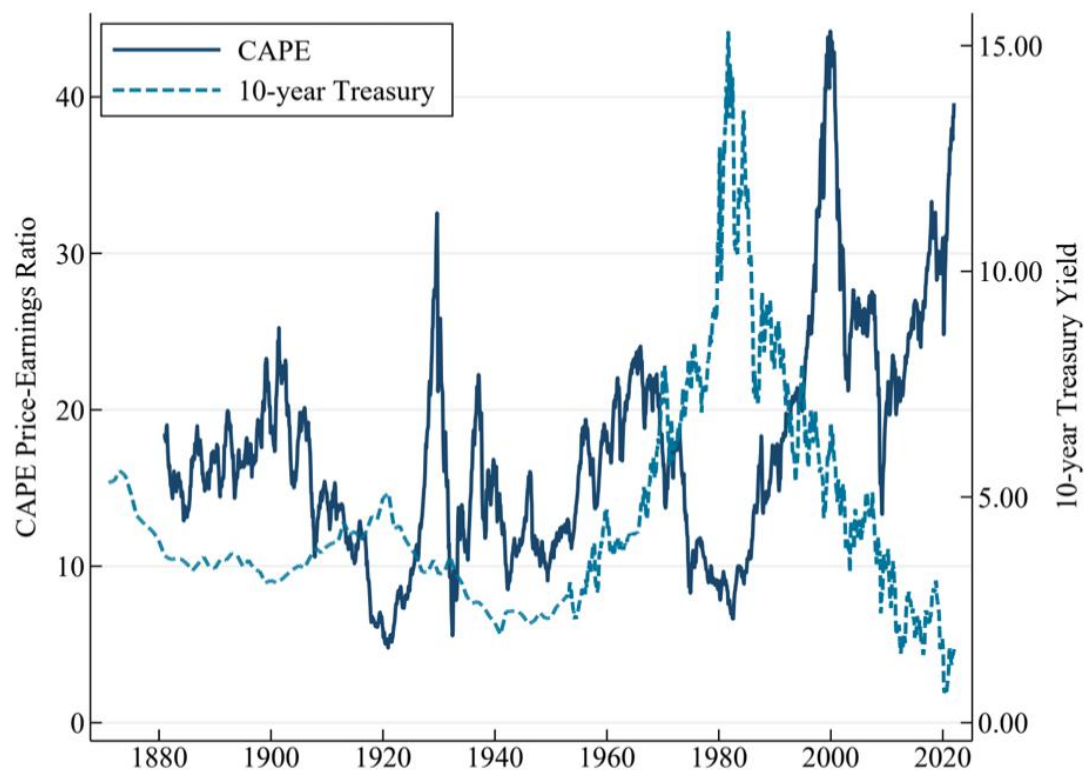
Valuation Ratios — Example

- ▶ A firm expects to earn $E_1 = \$4$ per share next year and pays out 60% of it as dividends
- ▶ Its dividends grow at 4% a year for ever, and its required return is 9%
- ▶ What is the share worth?
- ▶ What are its forward P/E and its forward P/D?
- ▶ What P/E would the market pay if growth were 5% instead of 4%?

Valuation Ratios — Answer

- ▶ **$D_1 = 0.60 \times 4 = \$2.40$, so $V_0 = 2.40 / (0.09 - 0.04) = \48.00**
- ▶ **Forward P/E = $48 / 4 = 12.0$ and forward P/D = $48 / 2.40 = 20.0$**
 - ▶ check with the formula: $P_0 / E_1 = (1 - b) / (r - g) = 0.60 / 0.05 = 12.0$
- ▶ **With $g = 5\%$: $V_0 = 2.40 / 0.04 = \$60.00$, and the P/E rises to 15.0**
- ▶ A multiple is not a separate theory. It is Gordon, divided by earnings
 - ▶ a high P/E says high growth, low risk, or both
- ▶ Pricing a share with a peer's P/E quietly assumes the peer has the same r and the same g

Markets: the US P/E Since 1880



Where is it today? multpl.com/shiller-pe

Growth Analysis

- ▶ In GGM, we can derive that the growth is equal to
 $b \times ROE$
- ▶ Let us try to understand it with an example:
- ▶ A company has \$100mm in assets, all equity financed
- ▶ Return on Equity (ROE) is 15%
- ▶ Plowback ratio is 60%

- ▶ What is the growth rate?
- ▶ If r is 12.5%, what is the value of the company?
- ▶ Recalculate with plowback ratio of 0%

Growth Analysis — Answer

- ▶ **$g = b \times \text{ROE} = 0.60 \times 15\% = 9\%$**
 - ▶ the firm keeps 60% of \$15mm and earns 15% on it: \$1.35mm more next year, which is 9% of \$15mm
- ▶ **Dividends are 40% of \$15mm = \$6mm, so $V_0 = 6 / (0.125 - 0.09) = \171.4mm**
- ▶ **With $b = 0$ everything is paid out and $g = 0$: $V_0 = 15 / 0.125 = \$120.0\text{mm}$**
- ▶ Reinvesting adds \$51.4mm of value
 - ▶ every euro kept inside earns 15%, while shareholders only require 12.5% elsewhere
- ▶ Paying less today made the shareholders richer. That only works while $\text{ROE} > r$

Growth Policy and Valuation

- ▶ Whether growth increases value depends on relationship between return on equity (ROE) and the discount rate (r)
 - ▶ If ROE is larger, then growth increases value
 - ▶ If ROE is smaller, then growth decreases value
 - ▶ If they are the same, then value does not depend on growth rate
- ▶ Intuition: ROE is rate of return on assets in this firm, r is like rate of return on outside investments with similar risk profile
- ▶ Worth investing more in this firm if return exceeds alternatives

Same Firm, Three ROEs

- ▶ The same firm: \$100mm of equity, plowback $b = 60\%$, required return $r = 12.5\%$

ROE	E_1	$g = b \times \text{ROE}$	D_1	V_0 with growth	V_0 if $b = 0$	Growth adds
15.0%	\$15.0	9.0%	\$6.0	\$171.4	\$120.0	+\$51.4
12.5%	\$12.5	7.5%	\$5.0	\$100.0	\$100.0	\$0.0
10.0%	\$10.0	6.0%	\$4.0	\$61.5	\$80.0	-\$18.5

Faster growth is not better growth. At $\text{ROE} = r$ growth is worth nothing; below r it destroys value

The firm with $\text{ROE} < r$ should pay the money out and let shareholders invest it at r

Growth Opportunities

- ▶ If a firm pays out all of its earnings ($b = 0$), it will not grow ($g = 0$) and will be worth

$$V_0^{b=0} = \frac{E[EPS_1]}{r}$$

- ▶ The difference between the firm value and the no-growth firm value is called the PV of growth opportunities (PVGO):

$$V_0 = \frac{E[EPS_1]}{r} + PVGO$$

- ▶ PVGO is positive if $ROE > r$

What should the firm do if $ROE < r$?

Growth Opportunities

- ▶ Apple's forward EPS is \$8 per share. Its required return is 10%. Its profitability (ROE) is expected to be 12%.
- ▶ What would Apple be worth if it paid out all of its earnings?
- ▶ In practice, Apple retains 75% of its earnings. How much is Apple worth?
- ▶ What is the value of Apple's growth opportunities?

Growth Opportunities — Answer

- ▶ **No growth ($b = 0$): $V_0 = \text{EPS}_1 / r = 8 / 0.10 = \80**
- ▶ **Retaining 75%: $g = 0.75 \times 12\% = 9\%$ and $D_1 = 0.25 \times 8 = \$2$, so $V_0 = 2 / (0.10 - 0.09) = \200**
- ▶ **PVGO = $200 - 80 = \$120$: 60% of the price is growth opportunities**
- ▶ And it is fragile. If ROE is 11% instead of 12%, $g = 8.25\%$ and $V_0 = 2 / 0.0175 = \$114.29$
 - ▶ one point of ROE takes 43% off the price
- ▶ A growth stock lives in the $r - g$ denominator. That is why it moves so much on so little news

Multi-Stage Growth

- ▶ Growth does not have to be a perpetuity. Firms often grow at different rates at different stages of development
- ▶ Growth Stage: rapidly expanding sales, many new investment opportunities, low payout ratio
- ▶ Transition stage: profit margin reduced by competition, fewer investment opportunities, higher payout ratio
- ▶ Mature stage: earnings growth, average return on equity, payout ratio stabilizes for the remaining life of the firm

DDM Case 3: Two-stage DDM

- ▶ A company can grow exceptionally for a while, but at some point the company matures and its growth normalizes.
- ▶ Suppose that you estimate that a company's growth will reach its “long-run” level of g after, say, 3 years
- ▶ Then, in 3 years its price is given by:

$$P_3 = \frac{(1 + g)D_3}{r - g}$$

- ▶ Based on estimates of the next 3 years' dividends, what is the price today?

$$P_0 = \frac{E[D_1]}{1 + r} + \frac{E[D_2]}{(1 + r)^2} + \frac{E[D_3]}{(1 + r)^3} + \frac{(1 + g)D_3}{(1 + r)^3(r - g)}$$

- ▶ Instead of estimating future growth rate, one can estimate future P/D ratio, or P/E ratio and earnings retention ratio

Two Stage DDM

- ▶ Ex: A stock's trailing dividend is \$4. Dividends are expected to grow by 25% per year for the next 2 years. The growth rate after that will be 5%. The required return is 15%
- ▶ How much is the stock worth?

Two-Stage DDM — Answer

- ▶ **$D_1 = 4 \times 1.25 = \$5.00$, $D_2 = 5 \times 1.25 = \$6.25$, $D_3 = 6.25 \times 1.05 = \6.5625**
- ▶ **From year 2 on dividends grow at 5% for ever, so Gordon prices the share at year 2:**
 - ▶ $P_2 = D_3 / (r - g) = 6.5625 / (0.15 - 0.05) = \65.63
- ▶ **$V_0 = 5 / 1.15 + (6.25 + 65.63) / 1.15^2 = 4.35 + 54.35 = \58.70**
- ▶ The trap: Gordon fed with D_3 gives a price at year 2, so it is discounted two periods, not three
 - ▶ Gordon always gives the value one period before its first dividend
- ▶ About 85% of today's value is the year-2 price. The long-run g matters more than the fast years

Example: Two-stage DDM

- ▶ Suppose you estimate:

Year	DPS
1	2.92
2	3.21
3	3.53
4	3.88

- ▶ The required return is constant at 8.2%
- ▶ After year 4, the dividend growth rate will be 6.4%
- ▶ Using the two-stage DDM, what is the fundamental value?

Example: Two-stage DDM

- Firstly, let's find the present value of the stock today, with Gordon pricing it at year 4:

$$\begin{aligned} P_0 &= \frac{E[D_1]}{1+r} + \frac{E[D_2]}{(1+r)^2} + \frac{E[D_3]}{(1+r)^3} + \frac{E[D_4]}{(1+r)^4} + \frac{(1+g)D_4}{(1+r)^4(r-g)} \\ &= \frac{2.92}{(1.082)} + \frac{3.21}{(1.082)^2} + \frac{3.53}{(1.082)^3} + \frac{3.88}{(1.082)^4} \\ &\quad + \frac{(1.064) \times 3.88}{(1.082)^4 \times (0.082 - 0.064)} \\ &= 178.4 \end{aligned}$$

Summary

- ▶ Gordon rearranged: $r = D_1 / P_0 + g$ — the dividend yield plus the growth rate
- ▶ Multiples are Gordon in disguise: $P_0 / E_1 = (1 - b) / (r - g)$
- ▶ Growth comes from reinvestment: $g = b \times \text{ROE}$
- ▶ Reinvesting creates value only if $\text{ROE} > r$. At $\text{ROE} = r$ growth is worth nothing; below r it destroys value
- ▶ $V_0 = \text{EPS}_1 / r + \text{PVGO}$: the price of a growth stock is mostly the price of its future projects
- ▶ When growth will not last, value the fast years one by one and put Gordon at the end — one period before its first dividend
- ▶ **Next class: Capital Budgeting Rules (BMAE 5.1–5.3)**